CSCI 2322 Homework 1

> **General Instructions:**
>
> 1. This is NOT an assignment to submit on Blackboard.
> 2. Please complete all problems and be prepared to show your work in person during the review session on Feb 17 to receive credit.

Tuesday, Feb 17th 2026

1. Let j, s, and k denote the propositions that Jasmine, Samir, and Kanti attend, respectively. Express the given conditions using logical connectives.
   (1) If Jasmine attends, then Samir should not attend.
   (2) If Samir attends, then Kanti should be there.
   (3) Kanti will not attend unless Jasmine also does.

2. There are ______ rows in the truth table for the compound statement containing propositional variables p, q, and s.

3. A logical binary relation $\odot$ is defined by the truth table below. Which one of the following propositional expressions is logically equivalent to $p \vee q$? Use a truth table to prove your answer.

$$\neg q \odot \neg q,\ p \odot \neg q,\ \neg p \odot q,\ \neg p \odot \neg q$$

p	q	$p \odot q$
T	T	T
T	F	T
F	T	F
F	F	T

4. Show that $(p \to q) \vee (p \to r)$ and $p \to (q \vee r)$ are logically equivalent (do not use truth table).

5. Show that $(\neg p \wedge (p \vee q)) \to q$ is a tautology (do not use truth table).

6. Let $F(x)$ denote "x is my friend", and $P(x)$ denote "x is perfect". What is the logical translation of the statement "All of my friends are perfect."

7. Let domain $= fleegles, snurds, thingamabobs$, predicates $F(x)$: x is a feegle, $S(x)$: x is a snurd, and $T(x)$: x is a thingamabob. Translate the following English sentences to predicate logic.

   (1) "Some fleegles are thingamabobs."

   (2) "All fleegles are snurds."

(3) "No snurd is a thingamabob."

8. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.

   (1) $\forall x\,(x^2 \geq x)$

   (2) $\forall x\,(x > 0 \vee x < 0)$

   (3) $\forall x\,(x = 1)$

9. The following sentence is taken from the specification of a telephone system: "If the directory database is opened, then the monitor is put in a closed state, if the system is not in its initial state." This specification is hard to understand because it involves two conditional statements. Find an equivalent, easier-to-understand specification that involves disjunctions and negations but not conditional statements.